Customer Retention Analytics

Made by Manav Mittal

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
import sqlite3
```

```
from google.colab import files

uploaded = files.upload()
```

Choose Files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving customer_churn_db to customer_churn (1).db

```
import os

print(os.path.getsize("customer_churn.db"))
```

```
16384
```

```
conn=sqlite3.connect('customer_churn.db')

sql_query="""
           SELECT name
           FROM sqlite_master
           WHERE type='table';
"""

tables=pd.read_sql(sql_query,conn)

#To create dataframe for each table

for table_name in tables['name']:
    df=pd.read_sql(f"SELECT * FROM {table_name}",conn)
    globals()[f"df_{table_name}"]=df
    print(f"Created dataframe:df_{table_name}")

conn.close()
```

```
Created dataframe:df_db_customer
Created dataframe:df_db_subscription
Created dataframe:df_db_support
```

```
# Print table names and column names
conn = sqlite3.connect('customer_churn.db')

for table_name in tables['name']:
    print(f"\nTable Name: {table_name}")
    # Get column information
    columns_query = f"PRAGMA table_info({table_name});"
    columns = pd.read_sql(columns_query, conn)
    print("Columns:")
    print(columns['name'].tolist())

# Close connection
conn.close()
```

```
Table Name: db_customer
Columns:
['customerid', 'name', 'country', 'state', 'gender', 'dob', 'interests', 'pincode']

Table Name: db_subscription
Columns:
['customerid', 'subscription_start_date', 'subscription_type', 'renewal_date', 'plan_type', 'contract_type', 'cancellation_d

Table Name: db_support
Columns:
['customerid', 'complaint_date', 'escalations', 'csat_score', 'col_1', 'comment']
```

```
df_db_customer.head()
```

	customerid	name	country	state	gender	dob	interests	pincode
0	0002-ORFBO	keshav	India	Maharashtra	Male	1982-04-12 00:00:00	travel	None
1	0003-MKNFE	raghav	India	Karnataka	Male	1995-11-23 00:00:00	None	None
2	0004-TLHLJ	lalita	India	Delhi	Female	1978-02-15 00:00:00	movie	None
3	0011-IGKFF	mohan	India	Nagaland	Male	2001-08-30 00:00:00	None	None
4	0013-EXCHZ	mira	India	Delhi	Female	1990-05-05 00:00:00	drama	None

```
df_db_customer.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 8 columns):
 #   Column      Non-Null Count  Dtype 
---  ------      --------------  ----- 
 0   customerid  21 non-null     object
 1   name        21 non-null     object
 2   country     18 non-null     object
 3   state       21 non-null     object
 4   gender      21 non-null     object
 5   dob         21 non-null     object
 6   interests   4 non-null      object
 7   pincode     0 non-null      object
dtypes: object(8)
memory usage: 1.4+ KB
```

```
df_db_customer.tail()
```

	customerid	name	country	state	gender	dob	interests	pincode
16	0020-JDNXP	rikim	India	Meghalaya	Female	1994-08-19 00:00:00	None	None
17	0021-IKXGC	vishakha	India	Rajasthan	Female	2000-09-02 00:00:00	None	None
18	0022-TCJCI	raghvendra	India	Telangana	Male	1983-12-30 00:00:00	None	None
19	0023-HGHWL	rishabh	India	Uttar Pradesh	Men	1991-05-14 00:00:00	None	None
20	0023-UYUPN	sudevi	India	Maharashtra	Women	1977-10-06 00:00:00	None	None

```
# rename col- name

df_db_customer.rename(columns={'name' : 'customer_name' },inplace=True)
```

```
df_db_customer.drop(columns=['interests','pincode'],inplace=True)
```

```
df_db_customer.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   customerid     21 non-null     object
 1   customer_name  21 non-null     object
 2   country        18 non-null     object
 3   state          21 non-null     object
 4   gender         21 non-null     object
 5   dob            21 non-null     object
dtypes: object(6)
memory usage: 1.1+ KB
```

```
df_db_customer['dob']=pd.to_datetime(df_db_customer['dob'])
```

```
# data standardization- gender

df_db_customer['gender'].unique()
```

```
array(['Male', 'Female', 'Women', 'Men'], dtype=object)
```

```
df_db_customer['gender']=df_db_customer['gender'].replace({'Men':'Male', 'Women': 'Female'})
```

```
df_db_customer['gender'].unique()
```

```
array(['Male', 'Female'], dtype=object)
```

```
df_db_customer[df_db_customer['country'].isna()]
```

	customerid	customer_name	country	state	gender	dob
5	0013-MHZWF	durga	None	Delhi	Female	1988-12-10
8	0015-UOCOJ	maya	None	Kathmandu	Female	1985-07-07
12	0018-NYROU	chitra	None	Telangana	Female	2004-12-01

```
state_country_mapping=df_db_customer.dropna(subset=['country']).set_index('state')['country'].to_dict()

df_db_customer['country']=df_db_customer['country'].fillna(df_db_customer['state'].map(state_country_mapping))
```

```
df_db_customer[df_db_customer['country'].isna()]
```

	customerid	customer_name	country	state	gender	dob

```
df_db_subscription.head()
```

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract_type	cancellation_date	cancel
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	Annual	None	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	Annual	2024-09-10	Switch
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	Monthly	None	

```
df_db_subscription.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              21 non-null     object
 1   subscription_start_date  21 non-null     object
 2   subscription_type       21 non-null     object
 3   renewal_date            21 non-null     object
 4   plan_type               21 non-null     object
 5   contract_type           21 non-null     object
 6   cancellation_date       6 non-null      object
 7   cancellation_reason     6 non-null      object
 8   monthly_charges         21 non-null     float64
 9   cltv                    21 non-null     int64
 10  churn_score             21 non-null     int64
dtypes: float64(1), int64(2), object(8)
memory usage: 1.9+ KB
```

```
date_col=['subscription_start_date','renewal_date','cancellation_date']

df_db_subscription[date_col]=df_db_subscription[date_col].apply(pd.to_datetime)

df_db_subscription.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 21 entries, 0 to 20
Data columns (total 11 columns):
 #   Column                  Non-Null Count  Dtype
---  ------                  --------------  -----
 0   customerid              21 non-null     object
 1   subscription_start_date  21 non-null     datetime64[ns]
 2   subscription_type       21 non-null     object
 3   renewal_date            21 non-null     datetime64[ns]
 4   plan_type               21 non-null     object
```

```
 5   contract_type            21 non-null     object
 6   cancellation_date        6 non-null      datetime64[ns]
 7   cancellation_reason      6 non-null      object
 8   monthly_charges          21 non-null     float64
 9   cltv                     21 non-null     int64
 10  churn_score              21 non-null     int64
dtypes: datetime64[ns](3), float64(1), int64(2), object(5)
memory usage: 1.9+ KB
```

```
df_db_support.head()
```

	customerid	complaint_date	escalations	csat_score	col_1	comment
0	0003-MKNFE	2024-08-28 00:00:00	N	60	None	service issue
1	0003-MKNFE	2024-08-28 00:00:00	Y	10	None	demaned refund
2	0013-EXCHZ	2024-01-20 00:00:00	Y	20	None	None
3	0013-MHZWF	2025-03-18 00:00:00	N	90	None	guidance to renew
4	0013-SMEOE	2024-11-01 00:00:00	N	30	None	None

```
df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      object
 1   complaint_date  9 non-null      object
 2   escalations     9 non-null      object
 3   csat_score      9 non-null      int64
 4   col_1           0 non-null      object
 5   comment         4 non-null      object
dtypes: int64(1), object(5)
memory usage: 564.0+ bytes
```

```
df_db_support.drop(columns=['col_1','comment'],inplace=True)
```

```
df_db_support.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9 entries, 0 to 8
Data columns (total 4 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   customerid      9 non-null      object
 1   complaint_date  9 non-null      object
 2   escalations     9 non-null      object
 3   csat_score      9 non-null      int64
dtypes: int64(1), object(3)
memory usage: 420.0+ bytes
```

```
df_db_support['complaint_date']=pd.to_datetime(df_db_support['complaint_date'])
```

```
# feature engineering and data analysis
```

```
# create a new column using exisiting column
df_db_subscription['churn flag']= np.where(df_db_subscription['cancellation_date'].notna(),1,0)
```

```
df_db_subscription.head()
```

	customerid	subscription_start_date	subscription_type	renewal_date	plan_type	contract_type	cancellation_date	cancel
0	0002-ORFBO	2021-03-15	Refferal	2025-03-15	Standard	Annual	NaT	
1	0003-MKNFE	2020-08-01	Paid	2024-08-01	Premium	Annual	2024-09-10	Switch
2	0004-TLHLJ	2022-11-20	Organic	2025-11-20	Basic	Monthly	NaT	

```
# first fix support table then merge
df=(df_db_subscription
                    .merge(df_db_customer,on='customerid',how='left')
                    .merge(df_db_support,on='customerid',how='left'))
```

```
df.shape
```

```
(23, 20)
```

```
df_db_subscription['customerid'].nunique()
```

```
21
```

```
df_db_customer['customerid'].nunique()
```

```
21
```

```
df_db_support['customerid'].nunique()
```

```
7
```

```
df_db_support['customerid'].size
```

```
9
```

```
df_db_support
```

	customerid	complaint_date	escalations	csat_score
0	0003-MKNFE	2024-08-28	N	60
1	0003-MKNFE	2024-08-28	Y	10
2	0013-EXCHZ	2024-01-20	Y	20
3	0013-MHZWF	2025-03-18	N	90
4	0013-SMEOE	2024-11-01	N	30
5	0017-IUDMW	2024-04-10	Y	25
6	0019-EFAEP	2024-09-27	Y	30
7	0022-TCJCI	2024-09-13	Y	10
8	0022-TCJCI	2024-09-14	N	90

```
df_db_support['complaint_count']=df_db_support.groupby('customerid')['customerid'].transform('count')
```

```
df_db_support=df_db_support.sort_values('complaint_date').drop_duplicates('customerid',keep='last')
```

```
df=(df_db_subscription
                    .merge(df_db_customer,on='customerid',how='left')
                    .merge(df_db_support,on='customerid',how='left'))
```

```
df.shape
```

```
(21, 21)
```

```
df.to_csv('customer_churn_dddata.csv', index=False)
```

```
# Data Analysis
```

```
# Churn rate
```

```
df.columns
```

```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count'],
      dtype='object')
```

```
churn_rate=df['churn flag'].mean()*100
print("churn rate = ", round(churn_rate,2),"%")
```

```
churn rate =  28.57 %
```

```
# Retention rate
```

```
retention_rate= 100- churn_rate
print("retention rate = ", round(retention_rate,2),"%")
```

```
retention rate =  71.43 %
```

```
# churn by plan type
```

```
churn_by_plan=df.groupby('plan_type')['churn flag'].mean().mul(100).round(2).reset_index(name='churn_rate_per')
print(churn_by_plan)
```

```
  plan_type  churn_rate_per
0     Basic           60.00
1   Premium           14.29
2  Standard           22.22
```

```
# churn by state
```

```
# average revenue per customer

arpu=df['monthly_charges'].mean()
print("average revenue per customer = ",round(arpu,2))
```

```
average revenue per customer =  18.85
```

```
df.columns
```

```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count'],
      dtype='object')
```

```
# average  customer tenure
# count of days users has used our service: cancellation date else current date
today=pd.Timestamp.today()

df['tenure_days']=np.where(df['cancellation_date'].notna(),
                           (df['cancellation_date']-df['subscription_start_date'] ).dt.days,
                           (today-df['subscription_start_date']).dt.days
                           )
avg_tenure=df['tenure_days'].mean()
print("average customer tenure = ",round(avg_tenure,2),"days")
```

```
average customer tenure =  1493.0 days
```

```
# revenue at risk - revenue lost from churned users
revenue_at_risk=df.loc[df['churn flag']==1,'monthly_charges'].sum()
print("revenue at risk = ",round(revenue_at_risk,2))
```

```
revenue at risk =  73.94
```

```
# esclation rate
escalation_rate=(df['escalations']=='Y').mean()*100
print("escalation rate = ",round(escalation_rate,2),"%")
```

```
escalation rate =  19.05 %
```

```
#  Avg Complaint Per User
avg_complaints = df['complaint_count'].sum() / df['customerid'].nunique()
print("Avg Compliants Per User = ", round(avg_complaints, 2))
```

```
Avg Compliants Per User =  0.43
```

```
# Correlation Esclation vs Churn
df['escalations'] = np.where(df['escalations'] == 'Y', 1, 0) # encoding string to int type
corr_df = df[['escalations', 'churn flag']].dropna()

correlation = corr_df['escalations'].corr(df['churn flag'])
print("Correlation between esclation vs churn is = ", round(correlation,2))
```

```
Correlation between esclation vs churn is =  0.77
```

```
# 11. Create a column using existing col - Churn risk
conditions = [
        (df['churn_score'] < 50),
        (df['churn_score'] >= 50) & (df['churn_score'] < 70),
        (df['churn_score'] >= 70)
]

choices = ['low', 'med', 'high']

df['churn_risk'] = np.select(conditions, choices, default='unkown')
```

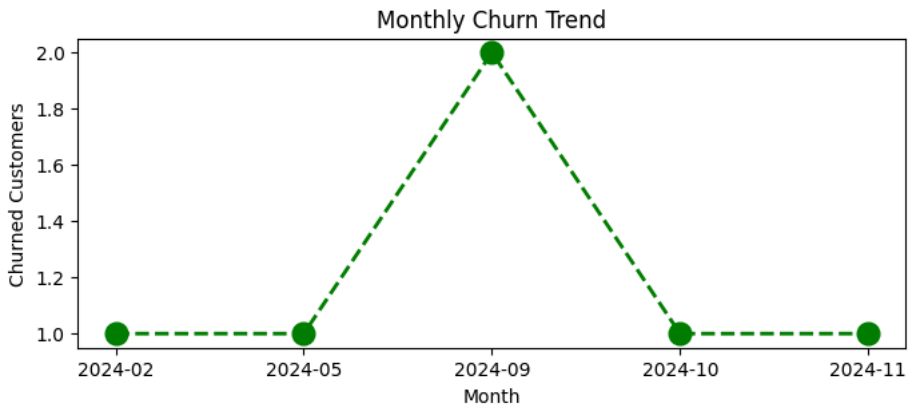
Visualization

```
# best practice to create a copy then work on it
df_visual = df.copy()
```

```
df_visual.columns
```

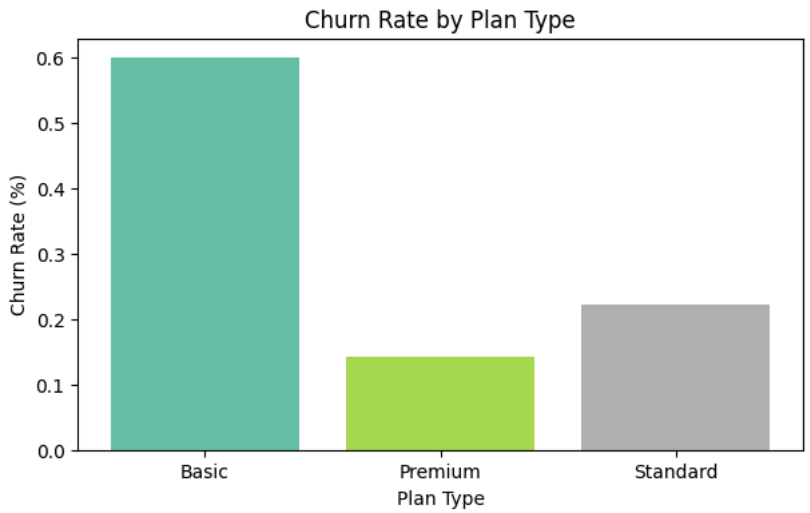
```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk'],
      dtype='object')
```

```
# 4.1 Monthly Churn Trend (Time Series KPI)

df_visual['cancellation_month'] = df_visual['cancellation_date'].dt.to_period('M')

churn_trend = df_visual[df_visual['churn flag'] == 1].groupby('cancellation_month').size()

plt.figure(figsize=(8,3))
plt.plot(churn_trend.index.astype(str), churn_trend.values,  color='green', marker='o', linestyle='dashed',  linewidth=2, r

plt.title('Monthly Churn Trend')
plt.xlabel('Month')
plt.ylabel('Churned Customers')
plt.show()
```

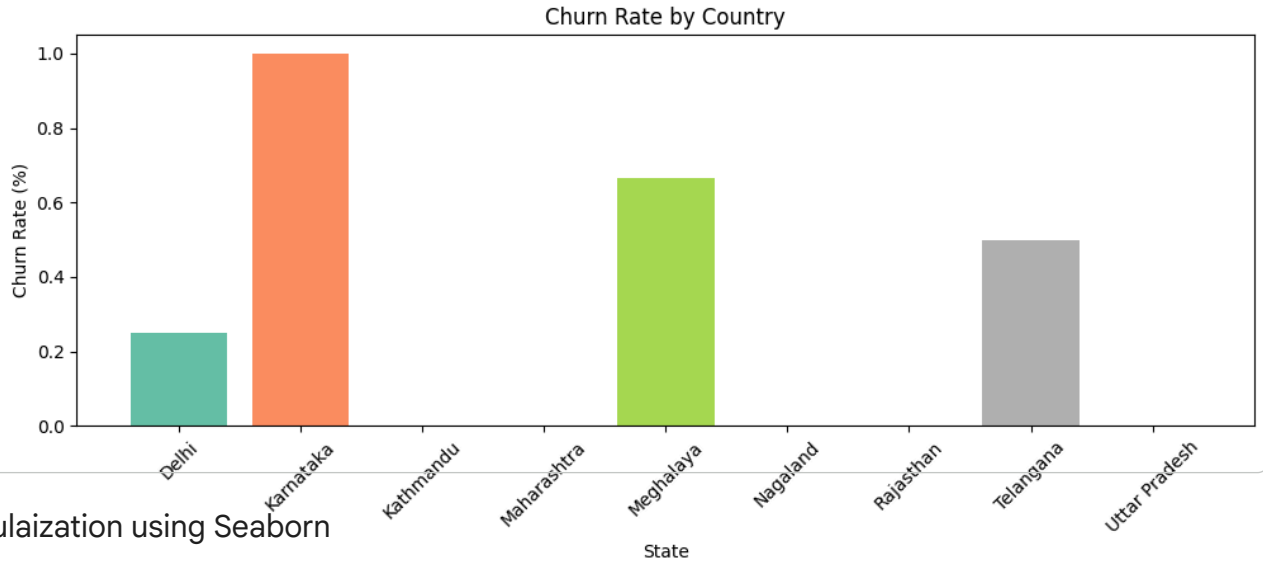



```
# 4.2 Churn Rate by Plan type
churn_plan = df_visual.groupby('plan_type')['churn flag'].mean()

# colors = ['yellow', 'purple', 'blue']
colors = plt.cm.Set2(np.linspace(0, 1, len(churn_plan)))

plt.figure(figsize=(7,4))
plt.bar(churn_plan.index, churn_plan.values, color = colors)

plt.title('Churn Rate by Plan Type')
plt.xlabel('Plan Type')
```

```
plt.ylabel('Churn Rate (%)')
plt.show()
```




```
# 4.3 Churn by States
churn_plan = df_visual.groupby('state')['churn flag'].mean()

# colors = ['yellow', 'purple', 'blue']
colors = plt.cm.Set2(np.linspace(0, 1, len(churn_plan)))

plt.figure(figsize=(12,4))
plt.bar(churn_plan.index, churn_plan.values, color = colors)

plt.title('Churn Rate by Country')
plt.xlabel('State')
plt.ylabel('Churn Rate (%)')
plt.xticks(rotation=45)
plt.show()
```



Visulaization using Seaborn

```
# Heatmap (Correlation Matrix)
```

```
# encoding - convert str to numeric so that we can find corr between features
df_visual.columns
```

```
Index(['customerid', 'subscription_start_date', 'subscription_type',
       'renewal_date', 'plan_type', 'contract_type', 'cancellation_date',
       'cancellation_reason', 'monthly_charges', 'cltv', 'churn_score',
       'churn flag', 'customer_name', 'country', 'state', 'gender', 'dob',
       'complaint_date', 'escalations', 'csat_score', 'complaint_count',
       'tenure_days', 'churn_risk', 'cancellation_month'],
      dtype='object')
```

```
df_visual[['plan_type', 'contract_type', 'churn_score', 'churn flag', 'churn_risk', 'escalations']].head()
```

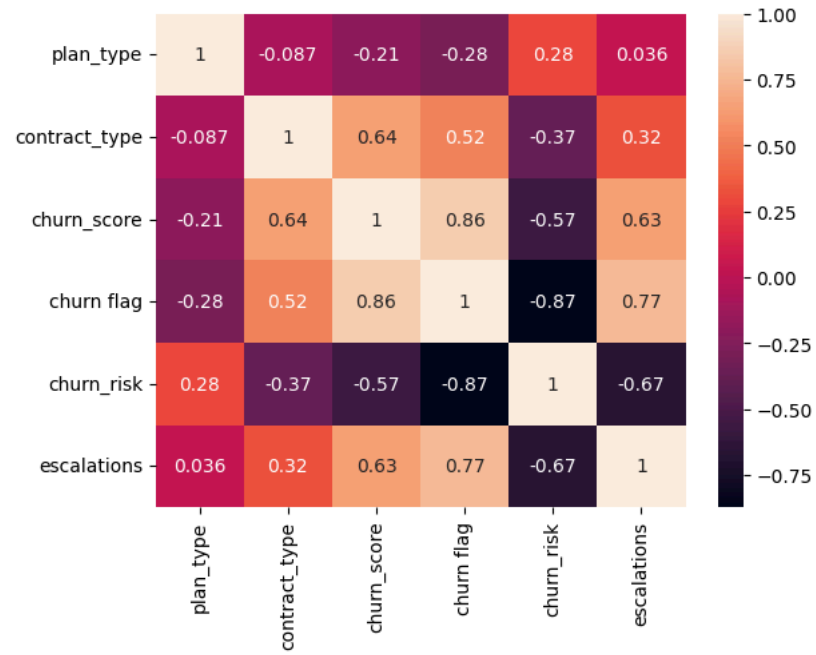
	plan_type	contract_type	churn_score	churn flag	churn_risk	escalations
0	Standard	Annual	12	0	low	0
1	Premium	Annual	91	1	high	1
2	Basic	Monthly	34	0	low	0
3	Premium	Annual	8	0	low	0
4	Standard	Monthly	88	1	high	1

```
# remove warnings
import warnings
warnings.filterwarnings("ignore")
```

```
# incorrect method of encoding - as numbers are not assigned based on priority
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn flag', 'churn_risk', 'escalations']]

categorial_cols = ['plan_type', 'contract_type', 'churn_risk']

for col in categorial_cols:
    df_encoded[col] = df_encoded[col].astype('category').cat.codes
```

```
# Heatmap (correlation matrix)

sns.heatmap(df_encoded.corr(), annot=True)
```

```
<Axes: >
```




```
print('plan_type :', df_visual['plan_type'].unique())
print('contract_type :', df_visual['contract_type'].unique())
print('churn_risk :', df_visual['churn_risk'].unique())
```

```
plan_type : ['Standard' 'Premium' 'Basic']
contract_type : ['Annual' 'Monthly']
churn_risk : ['low' 'high' 'med']
```

```
# Correct method of encoding - based on priority
df_encoded = df_visual[['plan_type', 'contract_type', 'churn_score', 'churn flag', 'churn_risk', 'escalations']]

order_mappings = {
    'plan_type' : ['Basic', 'Standard', 'Premium'],
    'contract_type' : ['Monthly', 'Annual'],
    'churn_risk': ['low', 'med', 'high']
    }

for col, order in order_mappings.items():
    df_encoded[col] = pd.Categorical(df_encoded[col].astype('category'), categories=order, ordered=True).codes
```

```
df_visual[['plan_type', 'contract_type', 'churn_score', 'churn flag', 'churn_risk', 'escalations']].head()
```

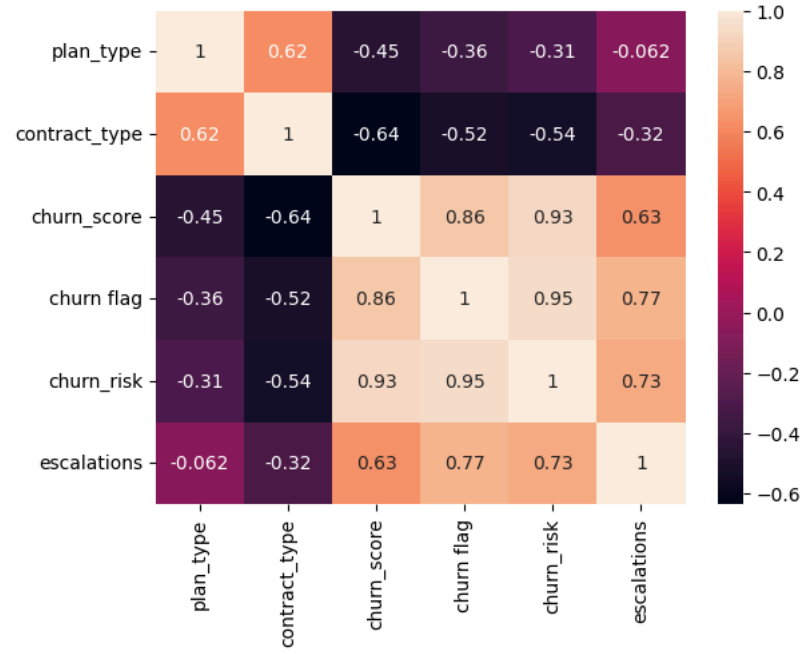
	plan_type	contract_type	churn_score	churn flag	churn_risk	escalations
0	Standard	Annual	12	0	low	0
1	Premium	Annual	91	1	high	1
2	Basic	Monthly	34	0	low	0
3	Premium	Annual	8	0	low	0
4	Standard	Monthly	88	1	high	1

```
df_encoded.head()
```

	plan_type	contract_type	churn_score	churn flag	churn_risk	escalations
0	1	1	12	0	0	0
1	2	1	91	1	2	1
2	0	0	34	0	0	0
3	2	1	8	0	0	0
4	1	0	88	1	2	1

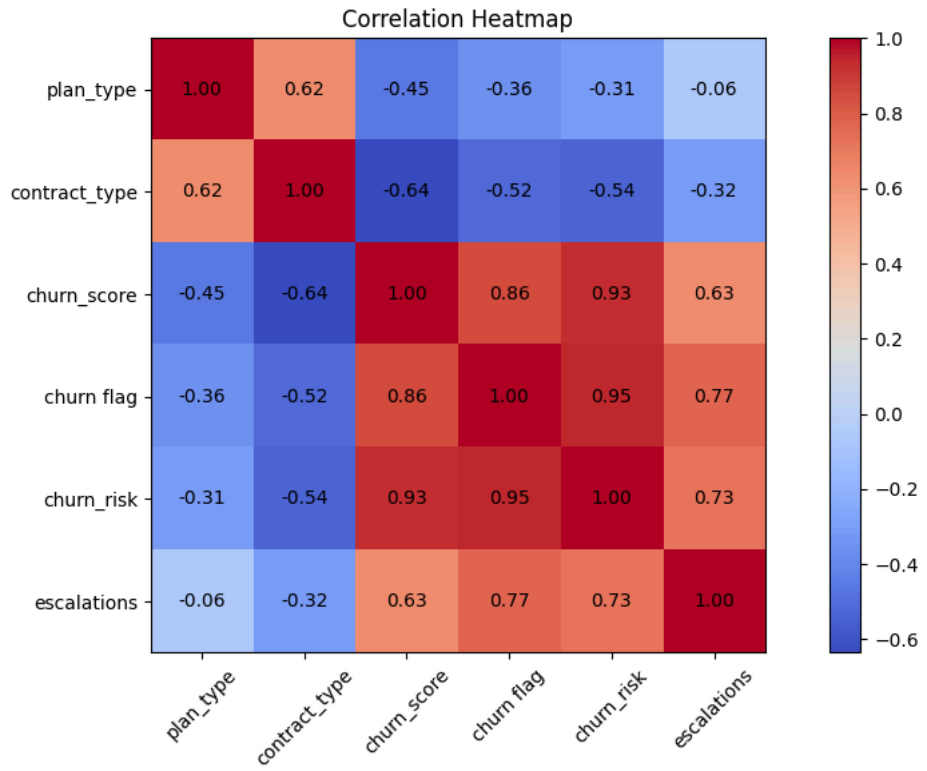
```
# Heatmap (correlation matrix)
sns.heatmap(df_encoded.corr(), annot=True)
```

```
<Axes: >
```



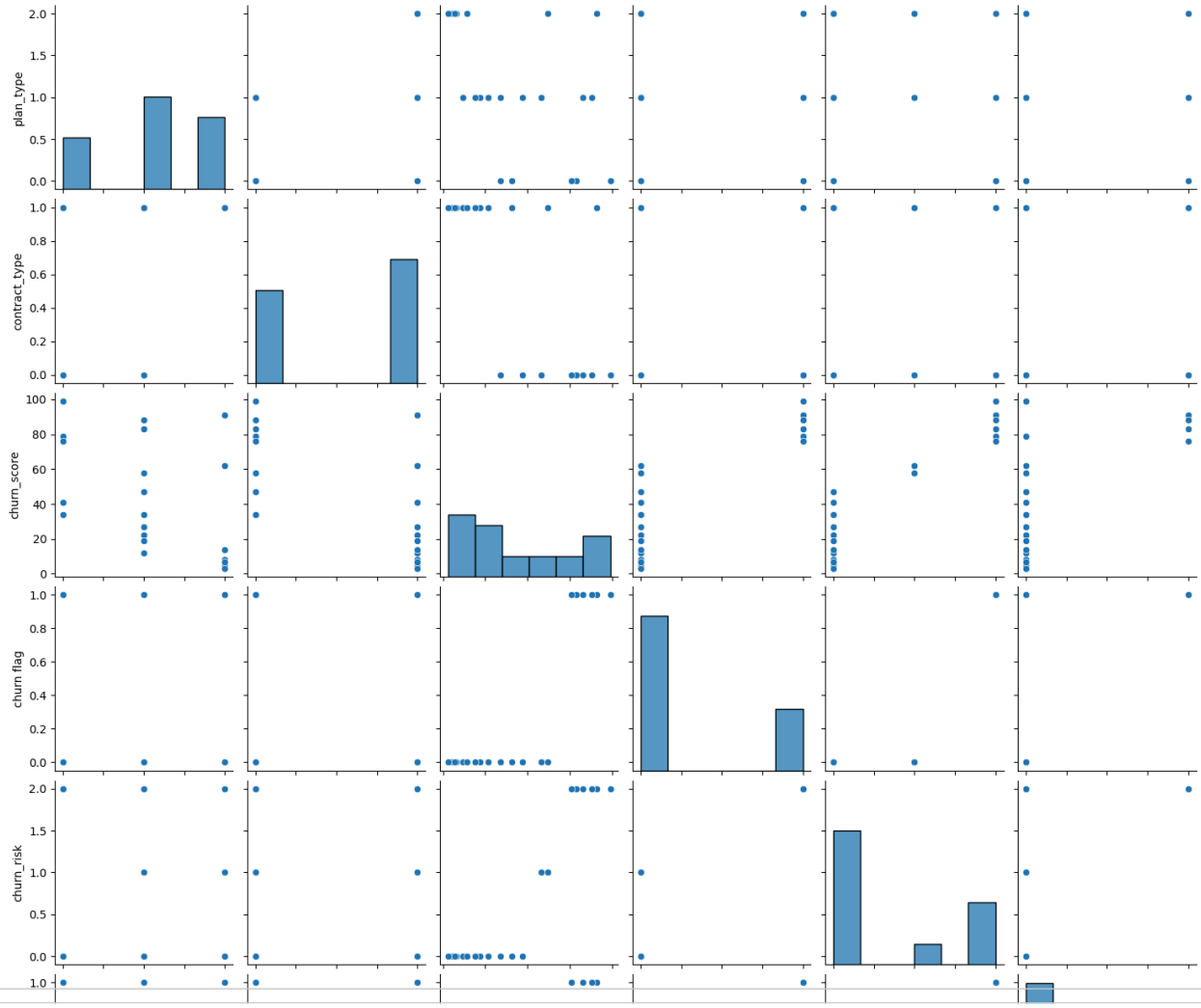

```
# Heatmap Using Matplotlib - difficult to plot it

corr_matrix = df_encoded.corr() # Correlation matrix

fig, ax = plt.subplots(figsize=(10, 6)) # Create figure

cax = ax.imshow(corr_matrix, cmap='coolwarm') # Heatmap

fig.colorbar(cax) # Add colorbar

# Axis labels
ax.set_xticks(np.arange(len(corr_matrix.columns)))
ax.set_yticks(np.arange(len(corr_matrix.columns)))

ax.set_xticklabels(corr_matrix.columns, rotation=45)
ax.set_yticklabels(corr_matrix.columns)

# Annotate values inside cells
for i in range(len(corr_matrix.columns)):
    for j in range(len(corr_matrix.columns)):
        ax.text(
            j,
            i,
            f"{corr_matrix.iloc[i, j]:.2f}",
            ha='center',
```

```
            va='center'
        )

plt.title('Correlation Heatmap') # Title

plt.tight_layout()
plt.show()
```



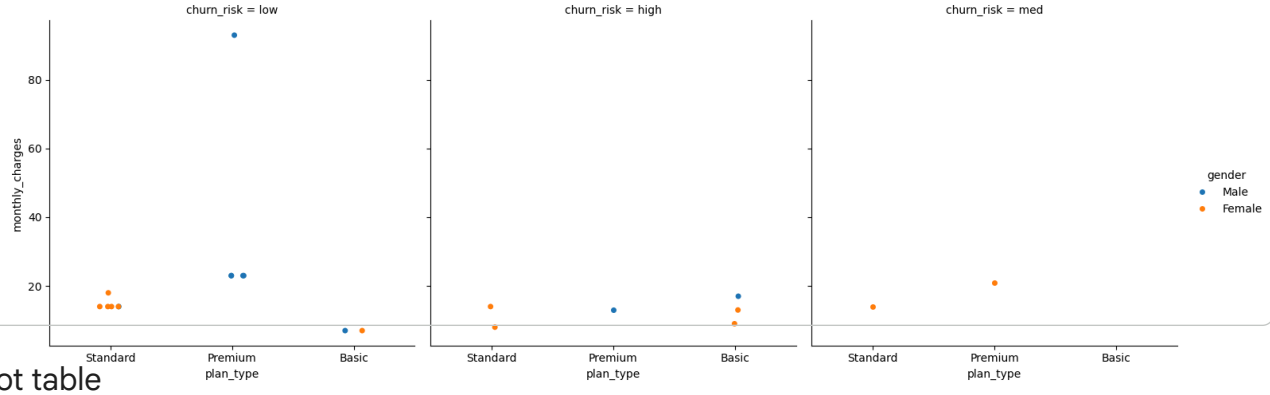

```
# pairplot - Plot pairwise relationships in a dataset
sns.pairplot(df_encoded)
```

```
<seaborn.axisgrid.PairGrid at 0x7ef84a435520>
```




```
# catplt/Facegrid plot - multi-dim comparison

sns.catplot(data=df_visual,
    x='plan_type',
    y='monthly_charges',
    hue='gender',
    col='churn_risk')
```

```
<seaborn.axisgrid.FacetGrid at 0x7ef84d8cc3e0>
```



Pivot table

```
# Pivot Table

pd.pivot_table(
    df_visual,
    index='plan_type',
    values='churn flag',
    aggfunc = 'mean'
)
```

plan_type	churn flag
Basic	0.600000

```
# pivot table using multiple cols and agg type

pd.pivot_table(
    df_visual,
    index='plan_type',
    values=['monthly_charges', 'customerid', 'churn flag'],
    aggfunc = {
        'monthly_charges' : 'sum',
        'customerid' : 'nunique',
        'churn flag' : 'mean'
    }
)
```

plan_type	churn flag	customerid	monthly_charges
Basic	0.600000	5	52.95
Premium	0.142857	7	218.93
Standard	0.222222	9	123.91

Working with sql and python

```
# create db in sql
conn = sqlite3.connect('test_database.sqlite')

#table details
conn.execute("CREATE TABLE users (first_name TEXT, country TEXT, budget INTEGER)")

# commit and save
conn.commit()

print("Cretaed Table successfully!")
```

```
Cretaed Table successfully!
```

```
# insert data

cursor = conn.cursor()

# write sql query to insert records in sql table
cursor.execute(
    """
        INSERT INTO users VALUES
            ('Madhav', 'India', 5000),
            ('Rishabh', 'Germany', 2500),
            ('Vishakha', 'India', 3500)
    """
)

# commit and save
conn.commit()

print("Data Inserted successfully!")
```

```
Data Inserted successfully!
```

```
# check inserted data in table
conn = sqlite3.connect('test_database.sqlite')
query = """SELECT * FROM users"""
```